Name **M AOUN**

DEPARTMENT **DATA SCIENCE AND AI**

LAB NO 1 :

Task 1: Print Name, Roll Number and Program

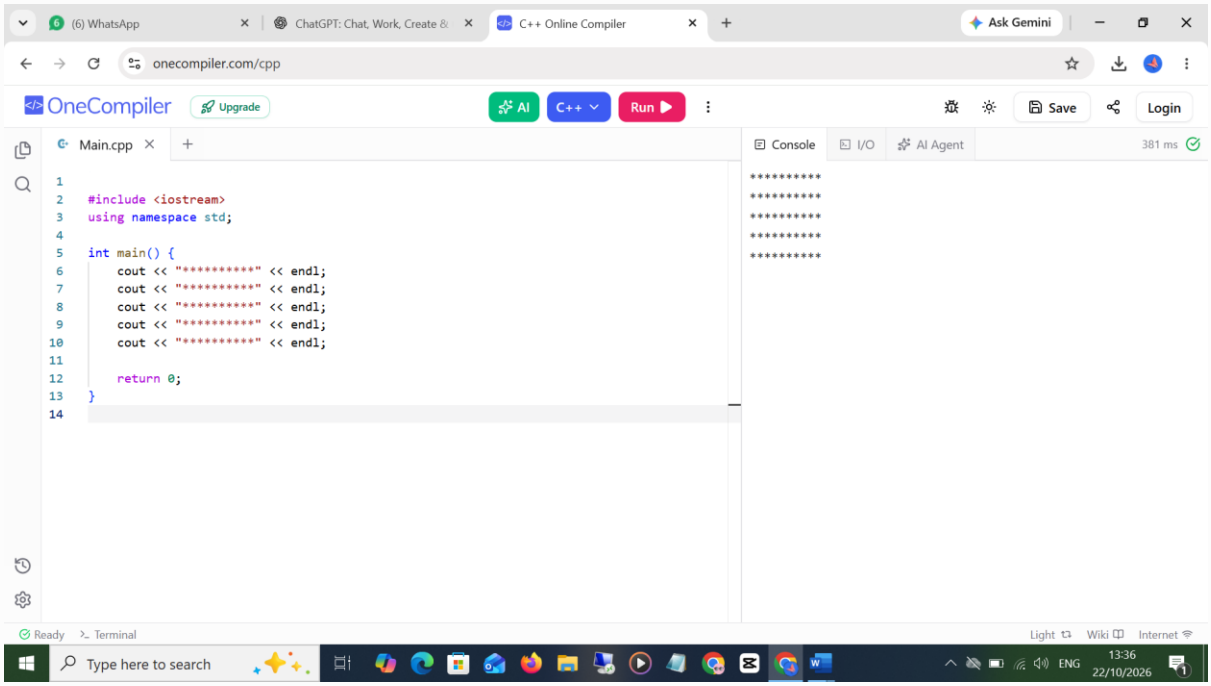
```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;
    cout << "Roll Number: Your Roll Number" << endl;
    cout << "Program: Your Department Name" << endl;

    return 0;
}
```

```
Name: M. Aoun
Roll Number: Your Roll Number
Program: Your Department Name
```

Task 2: Print a Box of Asterisks:



Task 3: Three Compiler Errors and Their Causes

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1. *You said:*

   is main hur ak program bnao and sepreat task and prhle pehla task and us main meraname m aoun and depart name ai

2. *ChatGPT said:*

   Bilkul! Main aap ke liye **C++ ka ek program** bana deta hoon jismein har lab task alag hoga. Pehle task mein aap ka name, roll number aur department alag-alag lines mein print hoga.

Task 1: Name, Roll Number aur DepartmenT

```
#include <iostream>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;
    cout << "Roll Number: Your Roll Number" << endl;
    cout << "Department: Your Department Name" << endl;

    return 0;
}
```

Output:

text

```
Name: M. Aoun
Roll Number: Your Roll Number
Department: Your Department Name
```

Apna actual roll number aur department name likh dein.

Task 2: Asterisks ka Box

```
#include <iostream>
using namespace std;

int main() {
    cout << "**********" << endl;
    cout << "**********" << endl;
    cout << "**********" << endl;
    cout << "**********" << endl;
    cout << "**********" << endl;

    return 0;
}
```

Output:

text

```
**********
**********
**********
**********
**********
```

Task 3: Three Compiler Errors

1. **Missing semicolon:** Statement ke end par ; na lagana.

2. **Undeclared variable:** Aise variable ko use karna jo declare nahi kiya gaya.
3. **Missing header file:** `#include <iostream>` na likhne ki wajah se `cout` ko pehchanne mein error aa sakta hai.

Task 4: GitHub par Repository Upload Karna

4. GitHub par account banayein ya login karein.
5. New repository create karein.
6. Apni C++ files upload ya push karein.
7. Repository ka link copy karke teacher ke saath share karein.

Aap mujhe apna roll number aur department ka naam bata dein, main Task 1 mein aapki actual details daal dunga.
Log in for personalized, step-by-step guidance.

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3. *You said:*

   JO TASK LIKO GE SAME TO SAME PAGE

4. *ChatGPT said:*

   Aap ke page par jo 4 lab tasks likhe hain, unko **same-to-same order mein**, har task ka alag C++ program bana kar likhna hai. Pehle task mein aapka naam **M. Aoun** aur department ka naam aayega.

Task 1: Print Name, Roll Number and Program

```
#include <iostream>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;
    cout << "Roll Number: Your Roll Number" << endl;
    cout << "Program: Your Department Name" << endl;

    return 0;
}
```

Task 2: Print a Box of Asterisks

```
#include <iostream>
using namespace std;
```

```
int main() {
    cout << "**********" << endl;
    cout << "**********" << endl;
    cout << "**********" << endl;
    cout << "**********" << endl;
    cout << "**********" << endl;

    return 0;
}
```

Task 3: Three Compiler Errors and Their Causes

```
#include <iostream>
using namespace std;

int main() {
    // Error 1: Missing semicolon
    // cout << "Hello"

    // Error 2: Undeclared variable
    // cout << number;

    // Error 3: Missing closing bracket
    // if (5 > 2) {
    //     cout << "Hello";

    return 0;
}
```

Explanation:

1. Missing semicolon: Statement ke end par semicolon `;` nahi lagaya.
2. Undeclared variable: `number` variable declare kiye baghair use kiya.
3. Missing closing bracket: `}` na lagane ki wajah se syntax error ho sakta hai.

LAB NO 2

Task 1: Area of Field in Square Metres and Kanals

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;

    double length, width, area, kanal;

    cout << "Enter length in metres: ";
    cin >> length;

    cout << "Enter width in metres: ";
    cin >> width;

    area = length * width;
    kanal = area / 505.857;

    cout << "Area in square metres: " << area << endl;
    cout << "Area in kanals: " << kanal << endl;

    return 0;
```

```
Name: M. Aoun
Enter length in metres: 67
Enter width in metres: 65
Area in square metres: 4355
Area in kanals: 8.60915
```

Task 2: Temperature Conversion into Fahrenheit and Kelvin

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;

    double celsius, fahrenheit, kelvin;

    cout << "Enter temperature in Celsius: ";
    cin >> celsius;

    fahrenheit = (celsius * 9.0 / 5.0) + 32;
    kelvin = celsius + 273.15;

    cout << fixed << setprecision(2);

    cout << "Temperature in Fahrenheit: "
         << fahrenheit << endl;

    cout << "Temperature in Kelvin: "
         << kelvin << endl;

    return 0;
}
```

```
Name: M. Aoun
Enter temperature in Celsius: 10
Temperature in Fahrenheit: 50.00
Temperature in Kelvin: 283.15
```

Task 3: Convert Seconds into Hours, Minutes and Seconds

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;

    int totalSeconds, hours, minutes, seconds;

    cout << "Enter total seconds: ";
    cin >> totalSeconds;

    hours = totalSeconds / 3600;
    minutes = (totalSeconds % 3600) / 60;
    seconds = totalSeconds % 60;

    cout << "Hours: " << hours << endl;
    cout << "Minutes: " << minutes << endl;
    cout << "Seconds: " << seconds << endl;

    return 0;
```

```
Enter total seconds: 1345
Hours: 0
Minutes: 22
Seconds: 25
```

Task 4: Demonstrate Overflow by Adding 1 to INT_MAX

```cpp
#include <iostream>
#include <climits>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;

    int maximum = INT_MAX;

    cout << "INT_MAX: " << maximum << endl;

    cout << "Adding 1 to INT_MAX causes "
         << "signed integer overflow." << endl;

    // Do not calculate maximum + 1:
    // Signed integer overflow in C++ is undefined behavior.

    return 0;
```

```
Name: M. Aoun
INT_MAX: 2147483647
Adding 1 to INT_MAX causes signed integer overflow.
```

LAB NO 3Task 1: Read Three Numbers and Print the Largest Using Nested If

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;

    int a, b, c;

    cout << "Enter three numbers: ";
    cin >> a >> b >> c;

    if (a > b) {
        if (a > c) {
            cout << "Largest number is: " << a << endl;
        } else {
            cout << "Largest number is: " << c << endl;
        }
    } else {
        if (b > c) {
            cout << "Largest number is: " << b << endl;
        } else {
            cout << "Largest number is: " << c << endl;
        }
    }
```

```
Name: M. Aoun
Enter three numbers: 12 34 56
Largest number is: 56
```

Task 2: Grade Calculator for Marks 0–100

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Name: M. Aoun" << endl;

    float marks;

    cout << "Enter marks (0-100): ";
    cin >> marks;

    if (marks < 0 || marks > 100) {
        cout << "Invalid marks!" << endl;
    }
    else if (marks >= 90) {
        cout << "Grade: A+" << endl;
    }
    else if (marks >= 80) {
        cout << "Grade: A" << endl;
    }
    else if (marks >= 70) {
        cout << "Grade: B" << endl;
    }
    else if (marks >= 60) {
```

```
Name: M. Aoun
Enter marks (0-100): 65
Grade: C
```

Task 3: Menu-Driven Calculator Using Switch

```cpp
#include <iostream>
using namespace std;
int main() {
    cout<<"Name: M. Aoun"<<endl;
    double a,b;
    char op;
    cout<<"Enter two numbers: ";
    cin>>a>>b;
    cout<<"Enter operator (+,-,*,/): ";
    cin>>op;
    switch(op) {
        case '+': cout<<a+b; break;
        case '-': cout<<a-b; break;
        case '*': cout<<a*b; break;
        case '/':
            if(b!=0) cout<<a/b;
            else cout<<"Division by zero!";
            break;
        default: cout<<"Invalid operator";
    }
    return 0;
```

```
Name: M. Aoun
Enter two numbers: 12 13
Enter operator (+,-,*,/): +
25
```

Task 4: Leap Year

```cpp
#include <iostream>
using namespace std;
int main() {
    cout<<"Name: M. Aoun"<<endl;
    int year;
    cout<<"Enter year: ";
    cin>>year;
    if(year%400==0) cout<<"Leap year";
    else if(year%100==0) cout<<"Not a leap year";
    else if(year%4==0) cout<<"Leap year";
    else cout<<"Not a leap year";
    return 0;
}
```

```
Name: M. Aoun
Enter year: 2026
Not a leap year
```